

EduVision_DV

Milestone 3: Tableau Dashboard Development

Modules 5 : University Overview & Research Analytics

Prepared as a practical Tableau worksheet blueprint. The structure below is based on the existing EduVision_DV implementation log, with the requested Rows / Columns / Marks / Filters instructions organized for direct dashboard construction.

1. Data Model Reference

The project uses a single unified Excel data model with a central relationship configuration. The documented logical model connects the four analytical tables, with 01 University Overview acting as the central university-level table.

Source Table	Target Table	Primary Key	Secondary Key	Relationship
01 University Overview	02 Research Analytics	University Id = University Id	Country Id = Country Id	Many-to-Many (Logical)
01 University Overview	03 Student Analytics	University Id = University Id	Country Id = Country Id	Many-to-Many (Logical)
01 University Overview	04 Country Comparison	Country Id = Country Id	None	Many-to-One (Lookup)

2. Module 5 : 01 University Overview Dashboard

Dashboard objective: present top university rankings, global university distribution, academic reputation, university performance movement, and institutional comparison.

2.1 Top University Rankings (Leaderboard Table)

Tableau Area	Configuration
Rows Shelf	rank_2024_numeric (Discrete / Blue Pill) university_name
Columns Shelf	Leave Blank
Marks Card	overall_score → Text (Average)
Filters Shelf	year = 2024; rank_2024_numeric = 1 to 10
Sorting / Axis	Sort rank_2024_numeric ascending.
Purpose / Note	Use rank_2024_numeric as the first row field so rank order is immediately visible.

2.2 Global University Distribution (Filled Map)

Tableau Area	Configuration
Rows Shelf	Latitude (generated)
Columns Shelf	Longitude (generated)
Marks Card	Country Name → Detail; University Id → Color (Count Distinct)
Filters Shelf	Country Name must have Geographic Role → Country/Region
Sorting / Axis	No numeric range filter required; optional year/country filters may be added at dashboard level.
Purpose / Note	Use a filled map to show the number of universities

	represented in each country.
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2.3 Academic Reputation Analysis (Institutional Leaderboard)

Tableau Area	Configuration
Rows Shelf	university_name
Columns Shelf	academic_reputation (Average)
Marks Card	Bar
Filters Shelf	rank_2024_numeric = 1 to 10
Sorting / Axis	Sort AVG(academic_reputation) descending.
Purpose / Note	Shows academic reputation among the highest-ranked universities.

2.4 University Performance Trends (Slope Chart)

Tableau Area	Configuration
Rows Shelf	Measure Values
Columns Shelf	Measure Names
Marks Card	Line; Measure Names → Path; university_name → Detail and Color
Filters Shelf	Measure Values contains only SUM(rank_2024_numeric) and SUM(rank_2025_numeric)
Sorting / Axis	Filter university_name to a benchmark cohort of 5–10 institutions.
Purpose / Note	Reverse the vertical axis so the best rank (1) appears at the top.

2.5 Institutional Comparison (Clustered Bar Chart)

Tableau Area	Configuration
Rows Shelf	university_name
Columns Shelf	Measure Values
Marks Card	Measure Names → Color; Measure Values contains AVG(academic_reputation) and AVG(overall_score)
Filters Shelf	Optional university selection / benchmark filter.
Sorting / Axis	Compare the two measures for the same institutions.
Purpose / Note	Use a grouped/clustered bar view for direct metric comparison.

3. Module 5 : 02 Research Analytics Dashboard

Dashboard objective: present research productivity, citation performance, research trends/comparisons, top research institutions, and research impact relationships. Use the actual fields available in 02 Research Analytics; do not invent publication counts where no publication-count field exists.

3.1 Research Productivity Analysis (Horizontal Bar Chart)

Tableau Area	Configuration
Rows Shelf	university_name (02 Research Analytics)
Columns Shelf	scores_research (Average)
Marks Card	Color → scores_research using a sequential gradient
Filters Shelf	rank_2024_numeric = 1 to 10
Sorting / Axis	Sort scores_research descending.
Purpose / Note	This is a research-performance view using the available Scores Research field.

3.2 Citation Performance vs. Research Productivity (Scatter Plot)

Tableau Area	Configuration
Rows Shelf	scores_citations (Average)
Columns Shelf	scores_research (Average)
Marks Card	Circle; university_name → Detail; Region → Color; opacity around 65%
Filters Shelf	rank_2024_numeric = 1 to 100
Sorting / Axis	Optional: uncheck Include Zero on both axes when necessary for distribution.
Purpose / Note	Interpret as research strength versus citation impact.

3.3 Research Productivity / Citation Comparison (Grouped Bar Chart)

Tableau Area	Configuration
Rows Shelf	university_name (02 Research Analytics) Measure Names
Columns Shelf	Measure Values
Marks Card	Measure Names → Color; Measure Values contains AVG(scores_research) and AVG(scores_citations)
Filters Shelf	rank_2024_numeric = 1 to 10
Sorting / Axis	Keep only the two research indicators.
Purpose / Note	Compare research and citation scores for the same top-ranked institutions.

3.4 Top Research Institutions (Heatmap Scorecard)

Tableau Area	Configuration
Rows Shelf	university_name (02 Research Analytics)
Columns Shelf	Measure Names
Marks Card	Square; Measure Values → Color and Text/Label
Filters Shelf	Measure Names limited to scores_citations, scores_industry_income, scores_research; rank_2024_numeric = 1 to 10
Sorting / Axis	Sort institutions by research score or a clearly documented benchmark.
Purpose / Note	Compact matrix comparing multiple research indicators.

3.5 Research Impact Comparison (Treemap)

Tableau Area	Configuration
Rows Shelf	No Rows field required
Columns Shelf	No Columns field required
Marks Card	Square; Size → AVG(scores_citations); Color → Region or Location; Label → Region + AVG(scores_citations)
Filters Shelf	Optional top-ranked or country filter.
Sorting / Axis	Use size for citation impact and color for geographic grouping.
Purpose / Note	Provides a geographic/block comparison of citation impact.

4. Field Placement Cheat Sheet

01 University Overview

Visualization	Rows	Columns	Key Marks / Filters
Top University Rankings	rank_2024_numeric;	—	overall_score → Text;

	university_name		year=2024; rank 1–10
Global University Distribution	Latitude (generated)	Longitude (generated)	Country Name → Detail; COUNTD(university_id) → Color
Academic Reputation	university_name	AVG(academic_reputation)	Bar; rank 1–10; descending reputation
Performance Trends	Measure Values	Measure Names	Line; Measure Names → Path; university_name → Detail + Color
Institutional Comparison	university_name	Measure Values	AVG(academic_reputation) + AVG(overall_score); Measure Names → Color

02 Research Analytics

Visualization	Rows	Columns	Key Marks / Filters
Research Performance	university_name	AVG(scores_research)	Color → scores_research; rank 1–10
Research vs Citations	AVG(scores_citations)	AVG(scores_research)	Circle; university_name → Detail; Region → Color; rank 1–100
Research + Citations	university_name; Measure Names	Measure Values	AVG(scores_research) + AVG(scores_citations); Measure Names → Color
Top Research Institutions	university_name	Measure Names	Square; Color + Label → Measure Values; rank 1–10
Research Impact Treemap	-	-	Size → AVG(scores_citations); Color → Region; Label → Region + citation score

5. Dashboard Layout Recommendation

Use a consistent executive-style layout: title and filters at the top, KPI/summary cards below the header, two primary analytical visuals in the center, and comparison/detail visuals in the lower section.

01 University Overview

- Header: University Overview + filters for Country, Region, University, and Year.
- First visual row: Top University Rankings + Global University Distribution.
- Second visual row: Academic Reputation + University Ranking Movement.
- Bottom row: Institutional Comparison.

02 Research Analytics

- Header: Research Analytics + filters for Country, University, and Rank.
- First visual row: Top Research Institutions + Citation Performance.
- Second visual row: Research vs Citation Scatter + Research/Citation Grouped Comparison.
- Bottom row: Research Impact Treemap / indicator scorecard.

6. Important Data & Visualization Notes

- 01 University Overview should retain rank_2024_numeric and rank_2025_numeric as the numeric ranking fields used for sorting and comparisons.

- Because a lower rank number represents better performance, ranking axes should be reversed where a trend/slope view is used so Rank 1 is visually at the top.
- O2 Research Analytics currently provides scores_research and scores_citations, but a publication-count field is not part of the documented field list. Do not create a publications chart unless a real publication field is added.
- A research 'trend' requires multiple years. If the current O2 Research Analytics dataset is only year 2024, present a cross-sectional comparison rather than a time-series trend.
- Use exact university_id / country_id relationships for data integration. Avoid filtering based on unrelated micro-level measures or accidental row-level fields.
- For dashboard actions, map Country Name → Country Name (or Location → Location where the target table uses Location) rather than passing specific score values.

7. Dashboard Interlinking

Action	Source	Target	Field Mapping	Run / Clear
Master Country Filter	World Map on Dashboard 1	All companion sheets on Dashboard 1	Country Name → Country Name	Run on Select; Clear → Show all values
Cross-Dashboard Country Filter	World Map on Dashboard 1	All worksheets on Dashboard 2	Location → Location (or Country Name → Country Name)	Use Selected Fields
Navigation	University Overview dashboard	Research Analytics dashboard	Navigation object	Text button: Go to Research Analytics →
Back Navigation	Research Analytics dashboard	University Overview dashboard	Navigation object	Text button: ← Back to University Overview

8. Source Reference

This document is based on the provided EduVision_DV Project Documentation & Implementation Log, which records the unified Excel model, relationship configuration, worksheet development settings, dashboard actions, and upcoming module roadmap.

The implementation log specifically documents the University Overview worksheet configurations and the Research Analytics worksheet configurations used as the structural basis for this dashboard blueprint.